SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Building Materials & Surveying: Stones, Cements, Leveling & Theodolite

Department: CIVIL	**Course / Subject:** Building Materials & Surveying (CE301PC)
Semester: Semester 3	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: Properties of construction stones, Cement types & hydration, Concrete mix, Chain surveying, Compass traversing, Leveling operations, and Total Station surveying.

Unit 1: Building Stones, Bricks & Timber

Igneous, sedimentary, metamorphic rocks, Compressive strength tests, Brick manufacturing (Kilns), Efflorescence test, Timber seasoning and defect prevention.

Unit 2: Cements, Aggregates & Mortar

Portland Pozzolana Cement (PPC), Bogue's compounds, Initial and final setting time test, Soundness test, Fineness modulus of aggregates, Mortar proportions.

Unit 3: Concrete Technology Fundamentals

Water-cement ratio, Slump test, Compacting factor test, Concrete compressive strength, Admixtures (Plasticizers, Retarders), Concrete curing methods.

Unit 4: Linear & Angular Surveying

Chain surveying methods, Chaining on slopes, Prismatic compass, Local attraction errors, Traverse computation, Coordinates and closing error adjustments.

Unit 5: Leveling, Contouring & Theodolite

Dumpy level, Height of Collimation method, Rise and Fall method, Curvature and refraction corrections, Contour intervals, and Transit Theodolite traversing.